LESSON 6.9

Divide Whole Numbers by Fractions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can explain what dividing a whole number by a fraction means — how many pieces fit, or how much is in one whole — find the quotient with a model or by multiplying by the reciprocal, and check it by multiplying.

WHY THIS MATTERS Agent Park has 4 pounds of evidence powder. Each testing bag holds exactly 1/2 pound. He needs to prepare as many testing bags as possible before the lab deadline. How many bags can he fill?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Whole Number *Spanish: Número entero*	A counting number with no fraction or decimal, like 0, 1, 2, 3.	4 is a whole number; as a fraction it is 4/1
Fraction *Spanish: Fracción*	A number that shows part of a whole, like 3/4.	1/2 means 1 out of 2 equal parts — like cutting a sandwich in half and taking one piece
Reciprocal *Spanish: Recíproco*	A fraction turned upside down.	The reciprocal of 2/3 is 3/2. Flip the fraction to divide: 6 ÷ 2/3 = 6 × 3/2
Keep, Change, Flip *Spanish: Mantener, cambiar, invertir*	A shortcut for dividing fractions: keep the first number, change ÷ to ×, flip the second. It works because dividing by a fraction gives the same answer as multiplying by its reciprocal.	6 ÷ 1/3 → Keep 6, Change ÷ to ×, Flip 1/3 to 3/1 → 6 × 3 = 18
Quotient *Spanish: Cociente*	The answer when you divide.	In 4 ÷ 1/2 = 8, the quotient is 8

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Agent Park has 4 pounds of powder and each bag holds 1/2 pound. I want 4 ÷ 1/2.

1. Each pound fills 2 half-pound bags, so 4 pounds fill 4 × 2 = 8 bags.
2. That is the same as multiplying by the reciprocal of 1/2: 4 × 2/1 = 8. Dividing by 1/2 doubles the count, because every whole holds 2 halves.
3. Check by multiplying: 8 bags × 1/2 pound = 4 pounds. So 4 ÷ 1/2 = 8. He can fill 8 bags.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's try 3 ÷ 1/4 together. How many 1/4s fit in 3? Each whole holds 4 fourths.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- 3 ÷ 1/4 = 12 because it takes _____ jumps of _____ to get from 0 to _____.

Word bank Whole Number | Fraction | Reciprocal | Keep, Change, Flip | Quotient | 12 | fourth | jumps | 3 | four times | each whole

WATCH OUT!

A common mistake

A common mistake in Divide Whole Numbers by Fractions is doing 'Keep, Change' but skipping the 'Flip'

LESSON 6.9

Divide Whole Numbers by Fractions

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

How many jumps of 1/4 does it take to get from 0 to 2?



HINT Each jump is exactly 1/4, and your goal is to land right on 2.

SHOW YOUR THINKING

PART 2 Practice the skill

show every step

2 **Circle the letter of the best answer. Show how you know.**

What is 6 ÷ 1/3?

(A) 2
(B) 3
(C) 6/3
(D) 18

HINT 6 ÷ 1/3 asks how many 1/3-size pieces fit into 6 wholes. Predict: bigger or smaller than 6?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is 3 ÷ 1/5?

(A) 3/5
(B) 5/3
(C) 8
(D) 15

HINT The total is 3 wholes and each piece is 1/5 — every whole holds several fifths.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is 10 ÷ 1/4?

(A) 2.5
(B) 10/4
(C) 14
(D) 40

HINT You're counting 1/4-size pieces inside 10 wholes — every whole holds more than one.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

What is 5 ÷ 1/2?

(A) 2.5
(B) 5/2
(C) 7
(D) 10

HINT How many 1/2-size pieces fit into 5 wholes? Each whole holds more than one half.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Problem** 4 ÷ 1/5
2. **Multiply by the reciprocal** Kept the 4 and changed ÷ to ×, but forgot to flip: 4 × 1/5
3. **Multiply** 4 × 1/5 = 4/5

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** After 'Change,' you still have to 'Flip' the second fraction.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does 3 ÷ 1/4 = 12? How does the number line show this?

SENTENCE STARTER 3 ÷ 1/4 = 12 because it takes ___ jumps of ___ to get from 0 to ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.9

Divide Whole Numbers by Fractions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

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WHY THIS MATTERS Agent Park has 4 pounds of evidence powder. Each testing bag holds exactly 1/2 pound. He needs to prepare as many testing bags as possible before the lab deadline. How many bags can he fill?

PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the table: Whole Number ÷ Fraction. Use the reciprocal shortcut.

Expression	Rewrite as Multiplication	Quotient
5 ÷ 1/4		
8 ÷ 1/2		
3 ÷ 1/6		
7 ÷ 1/3		

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each division expression to its quotient.

_____ 4 ÷ 1/3	(A) 12
_____ 7 ÷ 1/2	(B) 14
_____ 5 ÷ 1/6	(C) 30
_____ 2 ÷ 1/8	(D) 16
_____ 9 ÷ 1/4	(E) 36
_____ 6 ÷ 1/5	(F) 30

3 Use the model to solve. Show your work.



How many groups of 2/3 fit into 3?

SHOW YOUR WORK

Answer: _____

PART 2 Practice the skill

show every step

4 Circle the letter of the best answer. Show how you know.

What is 6 ÷ 2/3?

(A) 2/18
(B) 4
(C) 9
(D) 12

SHOW YOUR WORK

PART 3 Apply it

real-world problems

5 Circle the letter of the best answer. Show how you know.

Agent Park's 6 pounds of evidence powder fill 3/4 of a storage tub. How many pounds fill a WHOLE tub? (6 ÷ 3/4 asks how much is in one whole tub.)

(A) 4 1/2 pounds
(B) 8 pounds
(C) 2 pounds
(D) 6 3/4 pounds

SHOW YOUR WORK

PART 4 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Problem** 8 ÷ 3/4
2. **Multiply by the reciprocal** Flipped the wrong number — wrote 1/8 × 3/4 instead of 8 × 4/3
3. **Multiply** 1/8 × 3/4 = 3/32

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

Why does 3 ÷ 1/4 = 12? How does the number line show this?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 6.9

Divide Whole Numbers by Fractions

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Compare: what happens when you divide the same whole number by different fractions?

Expression	Quotient	Observation
12 ÷ 1/2		
12 ÷ 1/3		
12 ÷ 1/4		
12 ÷ 2/3		
12 ÷ 3/4		

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Division Error

1. **Problem** 6 ÷ 2/3
2. **Multiply by the reciprocal** 6 × 2/3
3. **Multiply** 12/3
4. **Simplify** 4

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A pizza shop has 8 whole pizzas. Each serving is 2/3 of a pizza. How many servings can they make? Write and solve an equation, then explain why the answer is greater than 8.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

Why does 3 ÷ 1/4 = 12? How does the number line show this?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help